

RIYA PATEL

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EDUCATION

B.Tech: Computer Science & Engineering, P P Savani University

2023 – 2027

Surat, Gujarat

Coursework: Data Structures & Algorithms · DBMS · Operating Systems · Computer Networks · OOP

SKILLS

- Python, Java, JavaScript (ES6+), HTML, CSS, SQL
- React, FastAPI, Spring Boot, Groq SDK, Docker, uvicorn, pytest, SpeechRecognition, psutil
- AWS (EC2, S3, RDS, IAM, VPC) · REST APIs · Microservices · Git & GitHub
- Agentic AI Systems · Async & Multi-threaded Programming · Chaos Engineering · TDD · NoSQL

PROJECTS

VoiceOS — AI Desktop Assistant

github.com/Riyadpatel24

Voice-controlled AI assistant with async execution, 3-tier security sandbox, and persistent memory.

- Designed an **agentic tool-use loop** with **ThreadPoolExecutor** — read-only tools run in parallel, state-mutating tools sequentially, keeping UI fully responsive during multi-step AI reasoning
- Engineered a **3-tier security sandbox** (SAFE / MODERATE / BLOCKED) using regex; file I/O locked to safe directories, system-critical commands permanently rejected at the guard layer
- **95%+ pytest coverage** across 30+ test cases; crash-safe persistent memory via atomic file replacement retains session history and tool logs across restarts
- **Exponential backoff** retry on rate limits and 5xx errors; integrated STT/TTS voice control, screenshot capture, and live system metrics

Adaptive System — Autonomous SRE Platform

github.com/Riyadpatel24/adaptive_system

Self-tuning platform with real-time telemetry, failure prediction, and chaos engineering.

- Built a **self-healing SRE loop**: telemetry ingestion → signal normalisation → health classification → LOCKDOWN/THROTTLE actions via safety guard and cooldown manager
- Failure prediction from rolling risk histories and root cause analysis via **DependencyGraph**; **PolicyEngine** adapts timeout and retry limits based on live failure rates
- **Chaos engineering** via CPU spike & memory leak fault injection; FastAPI daemon thread serves live snapshots at /snapshot for real-time observability

VoiceNotes Organizer — AI Audio-to-Notes Pipeline

github.com/Riyadpatel24/voicenotes-organizer

Full-stack pipeline: audio upload → Whisper transcription → LLM-structured notes, tasks, and decisions.

- React + FastAPI pipeline: audio transcribed via **Groq Whisper Large V3**, then structured by **LLaMA 3.3 70B** into summaries, action items, deadlines, and decisions in JSON
- **Modular architecture** cleanly separates transcription and structuring layers — AI models swappable without changing any downstream logic

CERTIFICATIONS

IBM Java Developer Professional Certificate — Coursera / IBM, Jun 2025

11 courses · Spring Boot, Microservices, Docker, REST APIs, NoSQL/SQL, Generative AI

[Verify certificate](#)

AWS Academy Graduate — Cloud Foundations — Amazon Web Services, Nov 2025

20 hours · EC2, S3, RDS, IAM, VPC, cloud security fundamentals

[Verify certificate](#)

LANGUAGES

- English: Professional (C1)
- Hindi: Native
- Gujarati: Native
- German: Elementary (A1)
- Mandarin Chinese: Elementary (A2)